

Module - 4

Filters

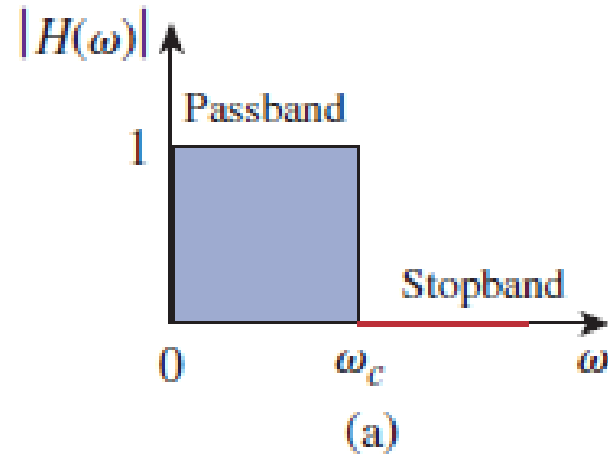
Dr. Sathya P
Associate Professor
SENSE

Filters

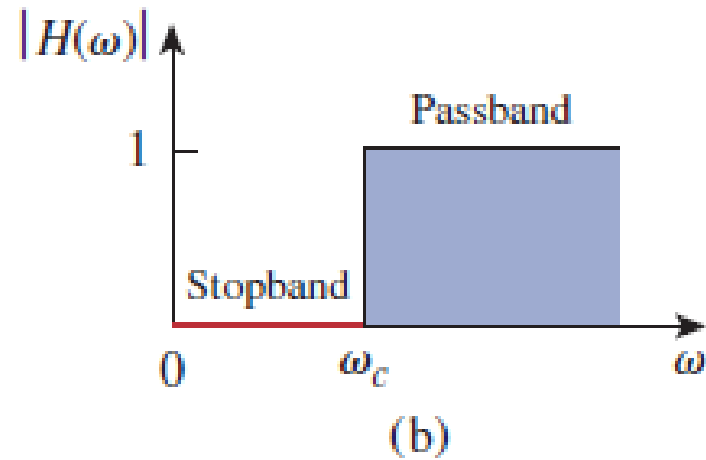
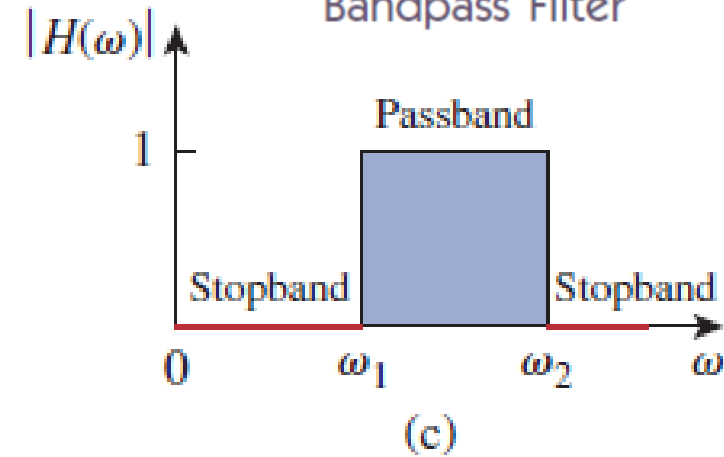
- A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.
- It is a frequency selective device, used to limit the frequency spectrum of a signal to some specified band of frequencies.
- They are used in radio and TV receivers to select one desired signal out of a multitude of broadcast signals in the environment.
- Filters can be active (Transistors, op-amps.,,) or passive (R, L, C).
- LC filters are used as equalizers, impedance matching networks, transformers, shaping networks, power dividers, attenuators, and directional couplers.

Types of Filters

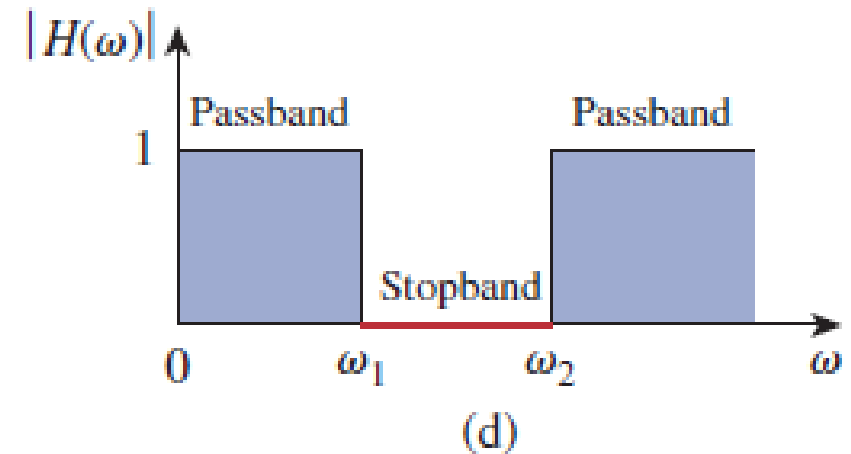
Lowpass Filter



Bandpass Filter



Highpass Filter



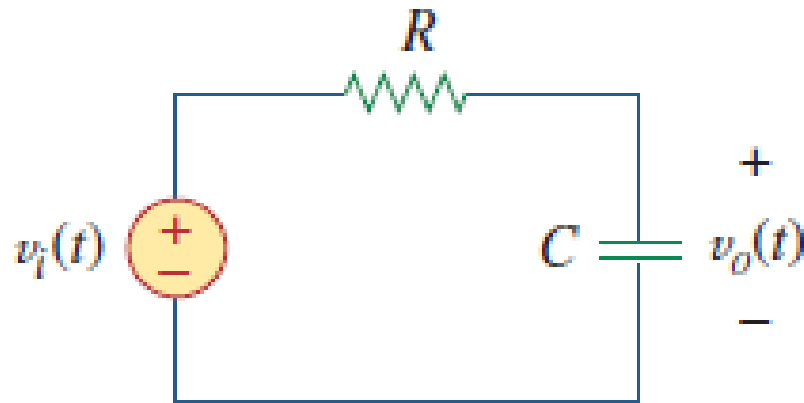
Bandstop Filter

Characteristics of ideal Filters

Type of Filter	$H(0)$	$H(\infty)$	$H(\omega_c)$ or $H(\omega_0)$
Lowpass	1	0	$1/\sqrt{2}$
Highpass	0	1	$1/\sqrt{2}$
Bandpass	0	0	1
Bandstop	1	1	0

Low Pass Filter

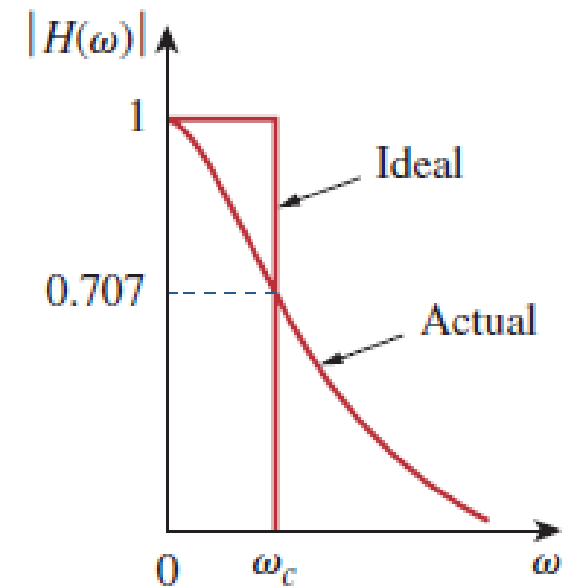
A **lowpass filter** is designed to pass only frequencies from dc up to the cutoff frequency ω_c .



Note:

For an RC LPF, the output is taken across the capacitor.

For an RL circuit, the output is taken across the resistor.



$$H(\omega) = \frac{V_o}{V_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

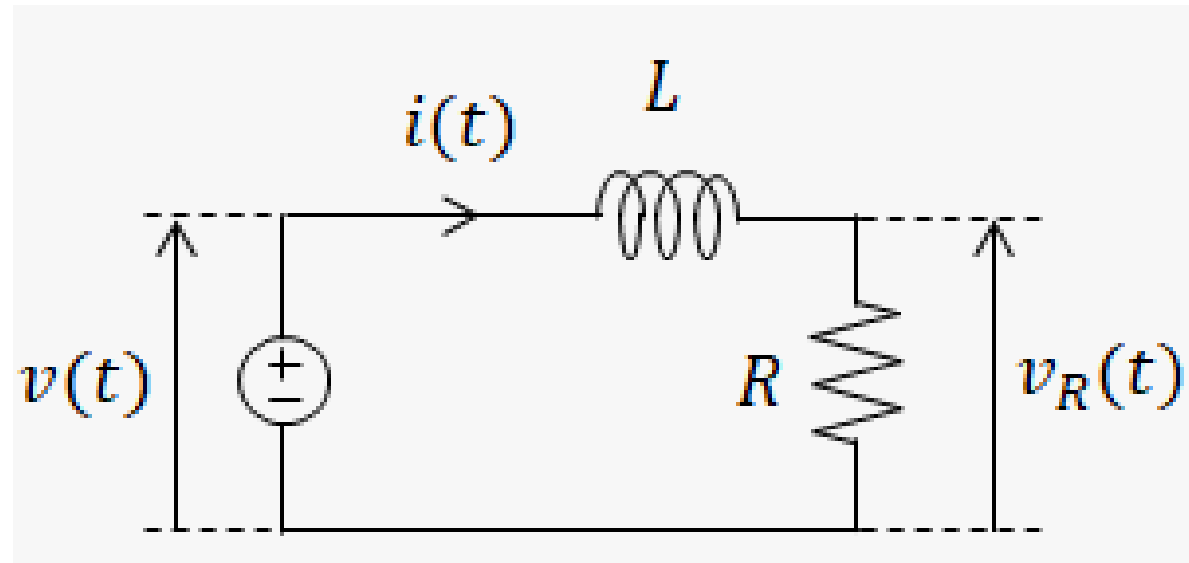
$$H(\omega) = \frac{1}{1 + j\omega RC}$$

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

$$\omega_c = \frac{1}{RC}$$

Numerical - 1

- Show that a series LR circuit is a low pass filter if the output is taken across the resistor. Calculate the corner frequency if $L = 2$ mH and $R = 10$ k Ω .



$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega L/R}$$

$H(0) = 1$ and $H(\infty) = 0$ showing that this circuit is a lowpass filter.

At the corner frequency, $|H(\omega_c)| = \frac{1}{\sqrt{2}}$, i.e.

$$\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \left(\frac{\omega_c L}{R}\right)^2}} \longrightarrow 1 = \frac{\omega_c L}{R} \quad \text{or} \quad \omega_c = \frac{R}{L}$$

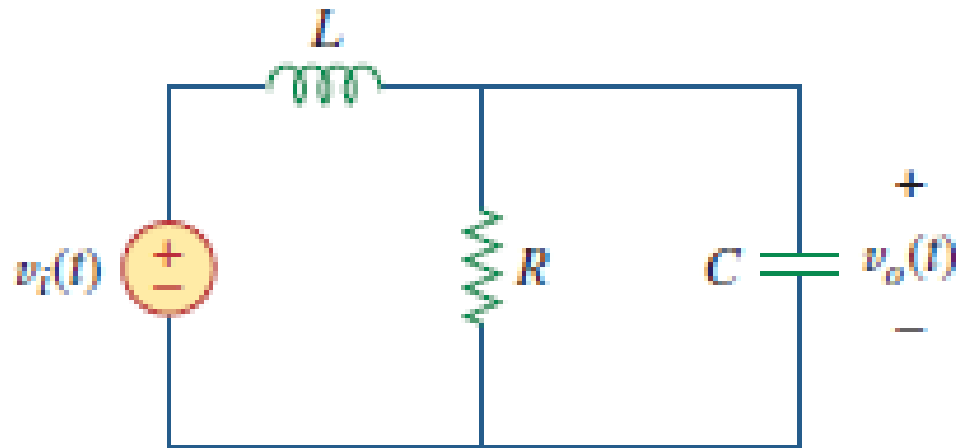
Hence,

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$f_c = \frac{1}{2\pi} \cdot \frac{R}{L} = \frac{1}{2\pi} \cdot \frac{10 \times 10^3}{2 \times 10^{-3}} = \underline{\underline{796 \text{ kHz}}}$$

Numerical - 2

- Find the transfer function of the circuit. Show that the circuit is a low pass filter.



- Solution:

$$H(\omega) = \frac{R \parallel \frac{1}{j\omega C}}{j\omega L + R \parallel \frac{1}{j\omega C}}$$

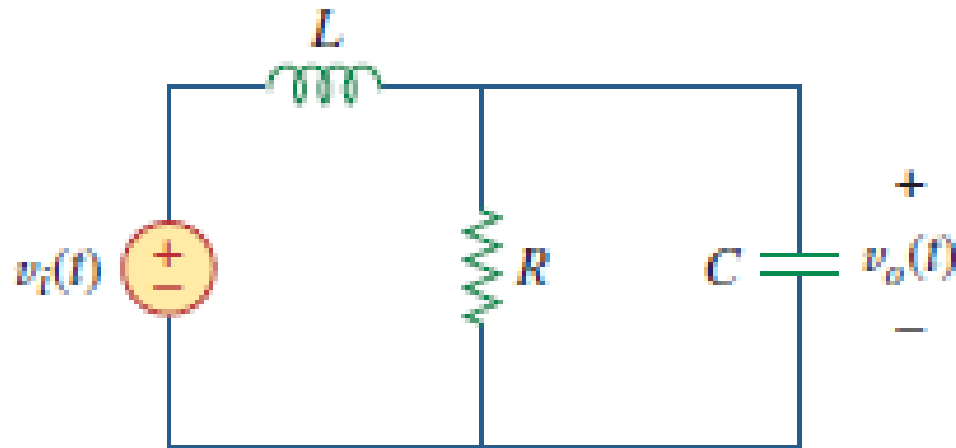
$$H(\omega) = \frac{\frac{R/j\omega C}{R + 1/j\omega C}}{j\omega L + \frac{R/j\omega C}{R + 1/j\omega C}}$$

$$H(\omega) = \frac{R}{\underline{R + j\omega L - \omega^2 RLC}}$$

$H(0) = 1$ and $H(\infty) = 0$ showing that this circuit is a lowpass filter.

Numerical - 3

- Determine what type of filter is shown in Figure. Calculate the corner or cut-off frequency. Take $R = 2 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 2 \text{ }\mu\text{F}$.



Solution:

The transfer function is

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R \parallel 1/sC}{sL + R \parallel 1/sC}, \quad s = j\omega$$

$$R \parallel \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

$$\mathbf{H}(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}, \quad s = j\omega$$

$$\mathbf{H}(\omega) = \frac{R}{-\omega^2RLC + j\omega L + R}$$

The corner frequency is the same as the half-power frequency, i.e., where $\mathbf{H}$ is reduced by a factor of $1/\sqrt{2}$. Since the dc value of $H(\omega)$ is 1, at the corner frequency,

$$H^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2 RLC)^2 + \omega_c^2 L^2}$$
$$2 = (1 - \omega_c^2 LC)^2 + \left(\frac{\omega_c L}{R}\right)^2$$

Substituting the values of R , L , and C , we obtain

$$2 = (1 - \omega_c^2 4 \times 10^{-6})^2 + (\omega_c 10^{-3})^2$$

Assuming that ω_c is in krad/s,

$$2 = (1 - 4\omega_c^2)^2 + \omega_c^2 \quad \text{or} \quad 16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

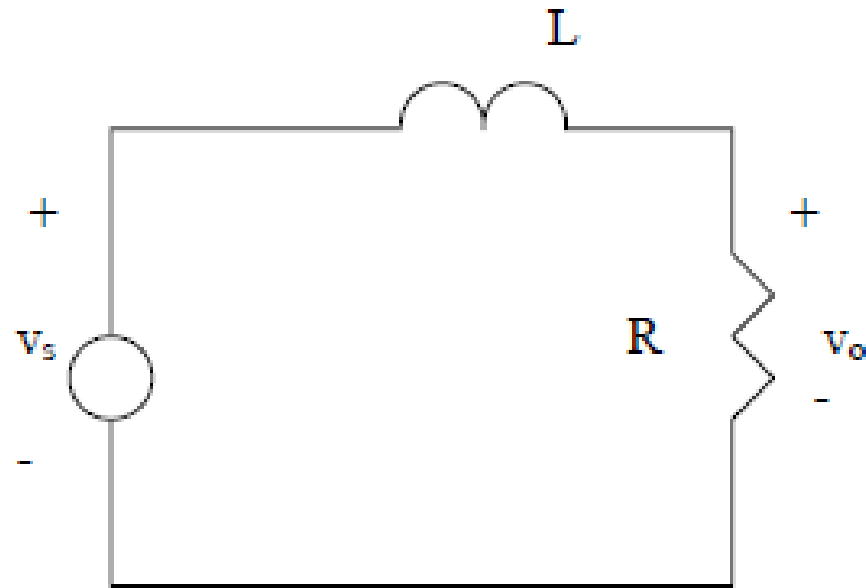
Solving the quadratic equation in ω_c^2 , we get $\omega_c^2 = 0.5509$ and -0.1134 . Since ω_c is real,

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Numerical - 4

- Design an RL low pass filter that uses a 40 mH coil and has a cut-off frequency of 5 kHz.

Solution:



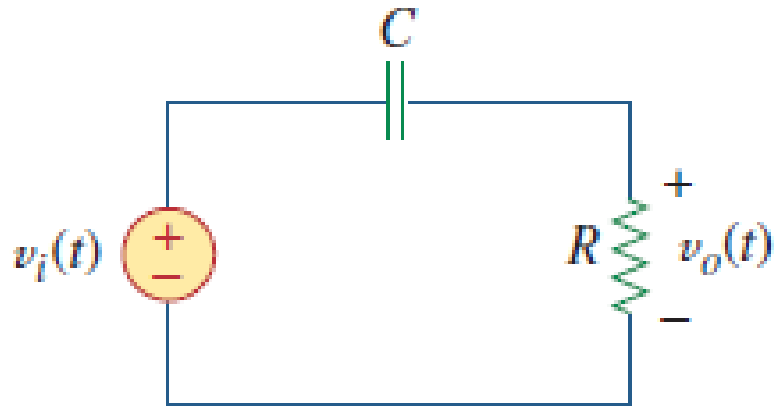
$$H = \frac{V_o}{V_s} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega L / R}$$

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$R = 2\pi f_c L = 2\pi \times 5 \times 10^3 \times 40 \times 10^{-3} = \underline{\underline{1.256 \text{ k}\Omega}}$$

High Pass Filter

A **highpass filter** is designed to pass all frequencies above its cutoff frequency ω_c .



$$H(\omega) = \frac{V_o}{V_i} = \frac{R}{R + 1/j\omega C}$$

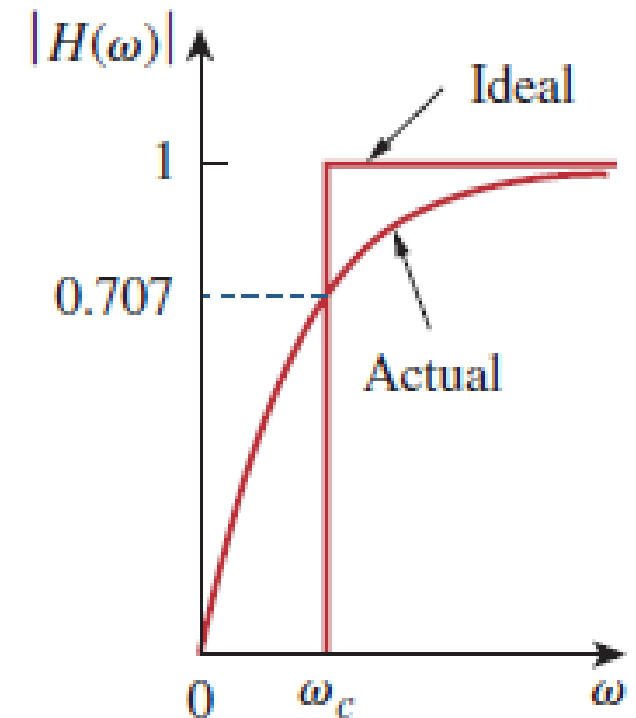
$$H(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

$$\omega_c = \frac{1}{RC}$$

Note:

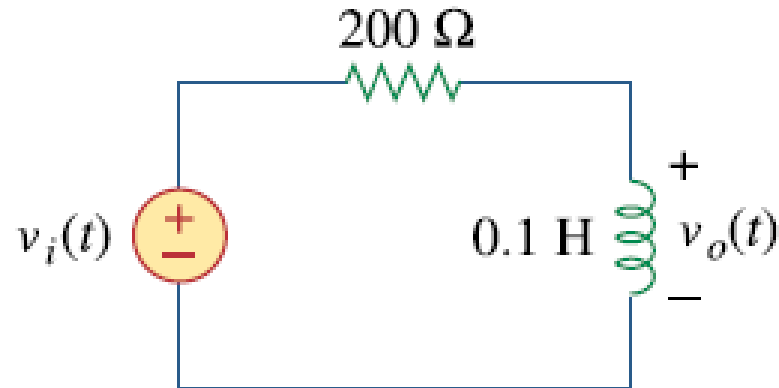
For an RC HPF, the output is taken across the resistor.

For an RL circuit, the output is taken across the inductor.



Numerical - 5

- Determine what type of filter is in Figure. Calculate the corner frequency in Hz.



- Solution:

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j\omega L}{R + j\omega L}$$

$H(0) = 0$ and $H(\infty) = 1$ showing that **this circuit is a highpass filter.**

$$\mathbf{H}(\omega_c) = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + \left(\frac{R}{\omega_c L}\right)^2}} \longrightarrow 1 = \frac{R}{\omega_c L}$$

or $\omega_c = \frac{R}{L} = 2\pi f_c$

$$f_c = \frac{1}{2\pi} \cdot \frac{R}{L} = \frac{1}{2\pi} \cdot \frac{200}{0.1} = \underline{\underline{318.3 \text{ Hz}}}$$

Numerical - 6

- In a highpass RL filter with a cutoff frequency of 100 kHz, $L = 40$ mH, find R.
- Solution:

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$R = 2\pi f_c L = (2\pi)(10^5)(40 \times 10^{-3}) = \underline{\underline{25.13 \text{ k}\Omega}}$$